

ME2171: Manufacturing Engineering

Semester: 4

Reverse engineering of Bicycle Air Pump

Project Report

By

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Summary

The reverse engineering project aims to dissect and analyse a handheld bicycle air pump, focusing on its constituent components. Through meticulous examination, we assess materials, manufacturing processes, and current design limitations. Our findings reveal opportunities for enhancement, including the utilization of alternative materials and shape modifications to increase durability and load-bearing capacity. By generating 3D models and proposing design changes, we contribute to the advancement of bicycle pump technology, aiming to create more efficient and user-friendly products. This project underscores the significance of continuous innovation in optimizing handheld bicycle air pumps for the benefit of cyclists worldwide.

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1 Introduction

The purpose of this project is to conduct a comprehensive reverse engineering analysis of a handheld bicycle air pump, with a focus on its constituent components. By dissecting and scrutinizing each part of the pump, we aim to gain insights into the materials used, manufacturing processes employed, potential drawbacks in current designs, and opportunities for enhancement.

Scope

Our scope encompasses a thorough examination of the materials utilized in key components of the bicycle air pump. We will delve into the manufacturing processes employed for each component, evaluating their efficiency, cost-effectiveness, and sustainability. Furthermore, we will identify any limitations or drawbacks associated with the current materials and propose alternative materials that may offer superior performance characteristics.

Objectives

1. Analyse the materials used in selected components of the bicycle air pump.
2. Evaluate the manufacturing processes employed for each component.
3. Identify drawbacks in the current materials and manufacturing methods.
4. Explore alternative materials that could enhance performance and durability.
5. Create 3D models of each component for further analysis and modification.
6. Propose design changes to optimize each components' load-bearing capacity and durability.

Methodology

Our methodology involves a systematic reverse engineering approach, beginning with the disassembly and documentation of the bicycle air pump. Each component will be meticulously examined to determine its material composition and manufacturing process. We will then assess the strengths and weaknesses of the current design, considering factors such as weight, hardness, durability, and efficiency. Utilizing advanced modelling software (Solid works), we will create 3D models of the components to facilitate further analysis and design modifications.

Selected components

Table 1: Selected components for analysing.

Pump barrel 	• serves as the compression chamber of the air pump. • Made with medium carbon steel • Made extrusion
Metal base 	• Provide counter force while in the upstroke which is an essential functionality. • Made with Gray cast iron • Sand casted
Handle 	• Used to apply the force to the metal rod in upstroke and downstroke • Made with medium carbon steel • Use rotary swaging to manufacture
Nozzle 	• Used to fix the hose into the bicycle type without any air leakage • Made with Polypropylene • Use injection moulding process to manufacture
Hose 	• Used to transfer air from pump barrel to the nozzle • Made with thermos plastic polyurethane • Use extrusion to manufacture

2 Task A – Material Identification

1. Pump barrel

Material Used	Medium carbon steel
Most likely manufacturing process	Extrusion

Standard or non-standard tests, and methods

Visual Identification Test

- Reasons: To get a rough idea about which tests should be performed on the specimen.
- Results: We decided to perform below mentioned tests on the specimen.

Performing Spark test

- Reasons: When tapping to the pump barrel, it gave us a metallic sound. Different metals produce different spark patterns when subjected to friction or grinding. By observing the colour, length, and intensity of the sparks, one can infer the type of metal present. This helps in confirming or refining the initial visual identification.
- Results: Gives an orange colour medium length spark and ends with a star which may have several rays.

Metallurgical Microscopic test

- Reasons: We got two possible material predictions (Medium Carbon Steel and Low Carbon Steel) that could've been used to produce the pump barrel by the Spark test. We did this test to further clarify which material is the exact material that used to manufacture the pump barrel.
- Results: Above grain structure shows a high similarity to the grain structure of Medium Carbon Steel

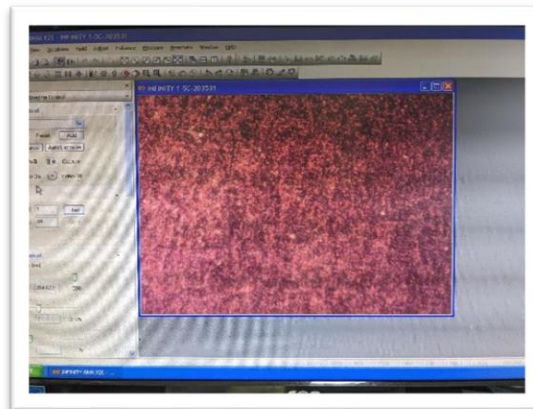


Figure 1: grain structure of Medium Carbon steel

Performing Hardness test

- Reasons: By conducting a Vickers hardness test, we can compare the obtained hardness value with standard hardness values for known materials. This comparison helps in confirming the material type and identifying any deviations from expected hardness levels, which could indicate variations in alloy composition or material quality.
- Results: Viker's hardness test value – 226.5 (Medium Carbon steel – 265 and Low Carbon Steel – 122.3)

2. Base

Material Used	Grey Cast Iron
Most likely manufacturing process	Sand Casting

Standard or non-standard tests, and methods

Visual Identification Test

- Reasons: To get a rough idea about which tests should be performed on the specimen. When tapping it gives a metallic sound.
- Results: We decided to perform below mentioned tests on the specimen.

Spark Test

- Reasons: Different metals produce different spark patterns when subjected to friction or grinding. By observing the color, length, and intensity of the sparks, one can infer the type of metal present. This helps in confirming or refining the initial visual identification.
- Results: Sparks travel in medium-length deep red ends with a tiny star. Assumed to be gray cast iron or white cast iron.

Metallurgical Microscopic test

- Reasons: We got two possible material predictions (Grey Cast Iron and White Cast Iron) that could've been used to produce the base by the Spark test. We did this test to further clarify which material is the exact material that used to manufacture.
- Results: Above grain structure shows a high similarity to the grain structure of Grey Cast Iron

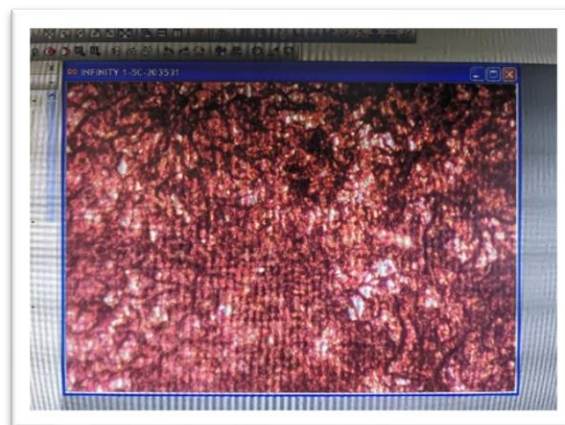


Figure 2: grain structure of metal base

Performing Hardness test

- Reasons: By conducting a Vickers hardness test, we can compare the obtained hardness value with standard hardness values for known materials. This comparison helps in confirming the material type and identifying any deviations from expected hardness levels, which could indicate variations in alloy composition or material quality.
- Results: Viker's hardness test value – 274 (Grey Cast Iron – 246 and White Cast Iron – 582)

3. Nozzle

Material Used	Polypropylene
Most likely manufacturing process	Injection Moulding

Standard or non-standard tests, and methods

Visual Identification Test

- Reasons: To get a rough idea about which tests should be performed.
- Results: We decided to perform below mentioned tests on the specimen

Performing FTIR test

- Reasons: By conducting an FTIR test, we can compare the obtained spectra with reference spectra of known materials. This comparison helps in confirming the material type and identifying any deviations or impurities present in the sample. For example, if the FTIR spectra match those of a specific polymer or plastic material, it provides confirmation that the nozzle is indeed made of that material.
- Results:
 1. Polymer classification – Polypropylene
 2. Trade name - Daplen DSC 1011
 3. Type of spectrum - TENSOR 27
 4. Source of spectrum - Bruker optik GMBH, Ettlingen, Germany
 5. Color – Black

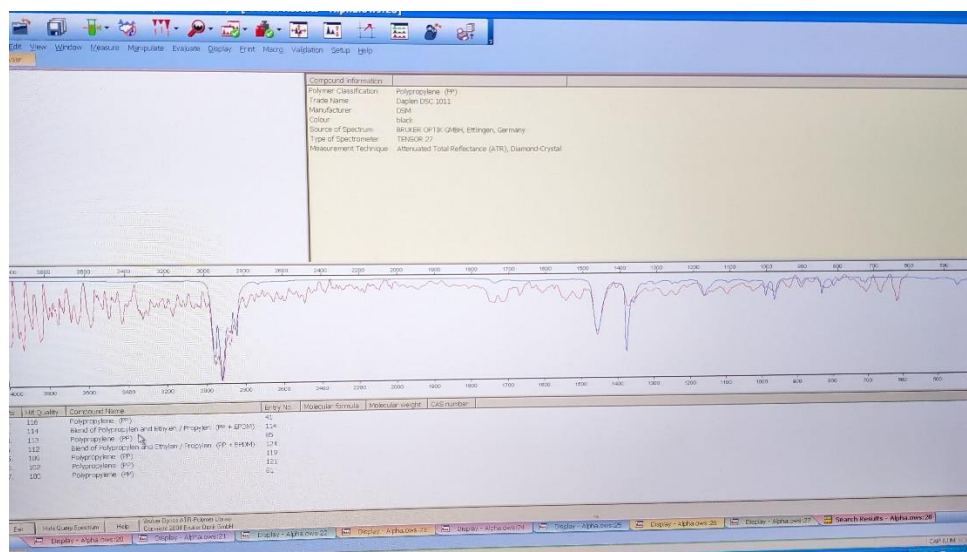


Figure 3: A photo of the results obtained by FTIR test

4. Handle

Material Used	Medium carbon steel
Most likely manufacturing process	Rotary Swaging

Standard or non-standard tests, and methods

Visual Identification Test

- Reasons: To get a rough idea about which tests should be performed on the specimen.
- Results: We decided to perform below mentioned tests on the specimen.

Performing Spark test

- Reasons: When tapping to the handle, it gave us a metallic sound. Different metals produce different spark patterns when subjected to friction or grinding. By observing the color, length, and intensity of the sparks, one can infer the type of metal present. This helps in confirming or refining the initial visual identification.
- Results: Gives an orange color medium length spark and ends with a star which may have several rays.

Metallurgical Microscopic test

- Reasons: We got two possible material predictions (Medium Carbon Steel and Low Carbon Steel) that could've been used to produce the handle by the Spark test. We did this test to further clarify which material is the exact material that used to manufacture the handle.
- Results: Below grain structure shows a high similarity to the grain structure of Medium Carbon Steel

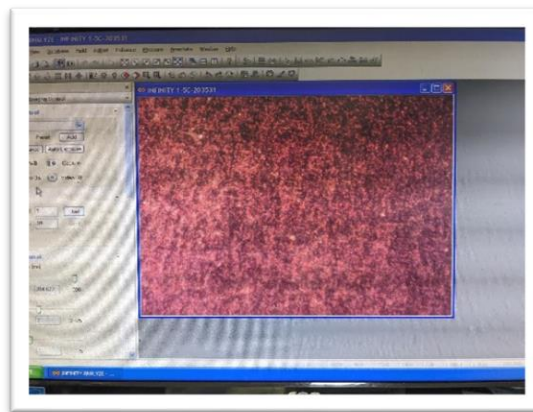


Figure 4: Grain structure of handle material

Performing Hardness test

- Reasons: By conducting a Vickers hardness test, we can compare the obtained hardness value with standard hardness values for known materials. This comparison helps in confirming the material type and identifying any deviations from expected hardness levels, which could indicate variations in alloy composition or material quality.

- Results: Viker's hardness test value – 226.5 (Medium Carbon steel – 265 and Low Carbon Steel – 122.3)

5. Exhaust Hose

Material Used	Thermo Plastic Polyurethane
Most likely manufacturing process	Extrusion

Standard or non-standard tests, and methods

Visual Identification Test

- Reasons: To get a rough idea about which tests should be performed on the specimen.
- Results: We decided to perform below mentioned tests on the specimen

Performing FTIR test

- Reasons: By conducting an FTIR test, we can compare the obtained spectra with reference spectra of known materials. This comparison helps in confirming the material type and identifying any deviations or impurities present in the sample. For example, if the FTIR spectra match those of a specific polymer or plastic material, it provides confirmation that the nozzle is indeed made of that material
- Results:
 - Polymer classification - Thermoplastic Polyurethane
 - Trade name - Desmopan KA 8377
 - Type of spectrum - TENSOR 27
 - Source of spectrum - Bruker optik GMBH, Ettlingen, Germany
 - Color - Nature
 - Density - 1200kg/m³
 - Melting temperature - 225C to 240C

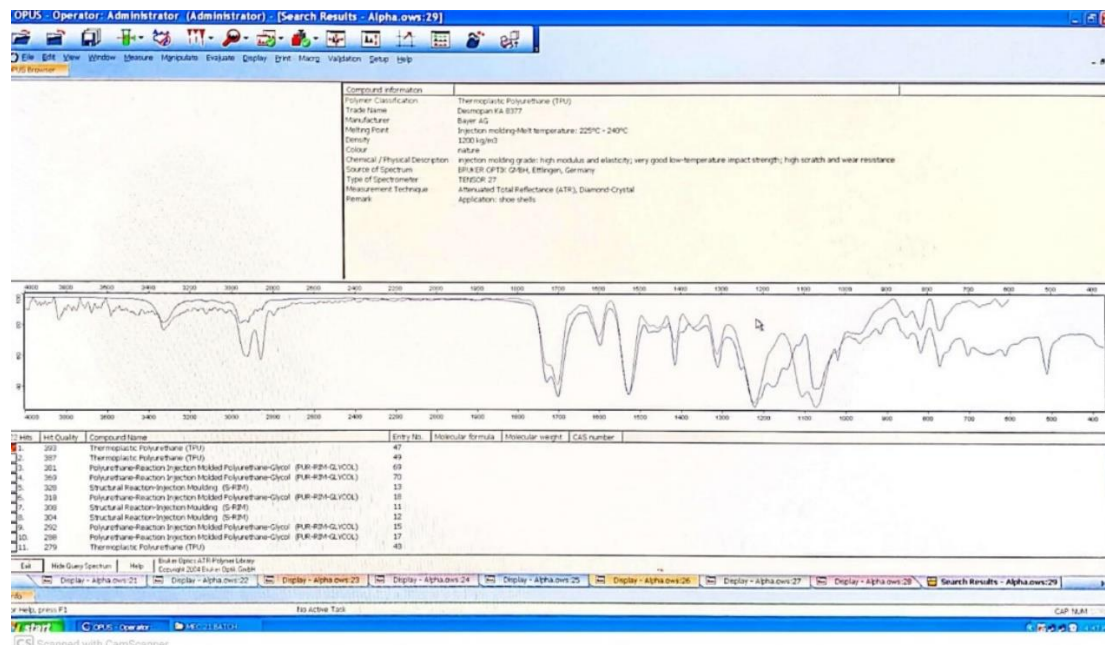


Figure 5: FTIR test results for exhaust hose

3 Task B – Discussing the casting possibilities

Pump Barrel

It is possible to manufacture the pump barrel by casting, but it might not be the most optimal method. Here are a few reasons:

1. **Material Properties:** Medium carbon steel typically requires controlled cooling rates but in casting, uneven cooling can be occurred.
2. **Complexity:** The shape and dimensions of a pump barrel specially the thickness is challenging to achieve in casting.
3. **Surface Finish:** Achieving a smooth surface finish which is required in pump barrel can't be done along with casting.
4. **Cost and Efficiency:** While casting can be cost-effective for certain components, it might not be the most efficient method for producing a relatively small and intricate part like a pump barrel.

Possible casting processes and the reasons for the selection

1. Sand Casting:
2. Investment Casting:
3. Die Casting:
4. Centrifugal Casting:

For the production of a pump barrel, sand casting is likely the most suitable casting process due to its versatility, cost-effectiveness, and ability to handle medium carbon steel while providing adequate dimensional accuracy and surface finish. Investment casting may also be considered for specific intricate parts, while die casting and centrifugal casting are less likely options for this particular component and material. Even if there are some casting processes that can be used to manufacture the pump barrel, casting is not used to manufacture it. Therefore, designing mold, pattern for this component will be unnecessary.

Base

Metal base of bicycle pump can be manufactured by casting. In fact, it is the most suitable method to manufacture the base. Here are the reasons:

1. **Suitable material properties:** Grey cast iron is well-suited for casting processes due to its fluidity when molten and its ability to fill intricate molds accurately.
2. **Cost-Effectiveness:** Casting is often a cost-effective manufacturing method for producing large quantities of components like the base of a bicycle pump. It allows for the production of complex shapes with relatively low tooling costs.
3. **Design Flexibility:** Casting allows for flexibility in design, enabling the creation of intricate shapes and features in the base of the pump to optimize its functionality and performance.

Possible changes to the material properties due to the moulding process

1. The cooling rate during casting influences the microstructure of grey cast iron, resulting in either finer-grained or coarser-grained structures, impacting properties like strength and machinability.
2. Porosity in cast iron arises from gas entrapment during solidification, decreasing mechanical properties such as tensile strength and fatigue resistance.
3. Residual stresses develop in cast components due to non-uniform cooling, affecting dimensional stability and mechanical behavior, potentially causing distortion or cracking under load.
4. Inclusions of impurities introduced during casting can diminish the mechanical properties of cast iron, including reducing ductility and toughness.
5. Variations in grain size throughout cast components due to differing cooling rates can affect properties like fatigue resistance and thermal conductivity.
6. Post-casting heat treatment processes can modify the microstructure and properties of grey cast iron, altering hardness, strength, and machinability based on specific treatment parameters.

Possible casting processes and the reasons for the selection

1. Sand Casting
2. Shell Moulding
3. Centrifugal Casting

The most suitable casting process for the metal base made is sand casting. Sand casting is chosen because it is highly versatile and widely used for casting Gray cast iron components. It offers flexibility in design, accommodates relatively simple to moderately complex shapes, and is cost-effective for producing large components like the base. Additionally, sand casting provides good surface finish and dimensional accuracy for Gray cast iron parts, making it an optimal choice for this application.

Required designs for manufacturing

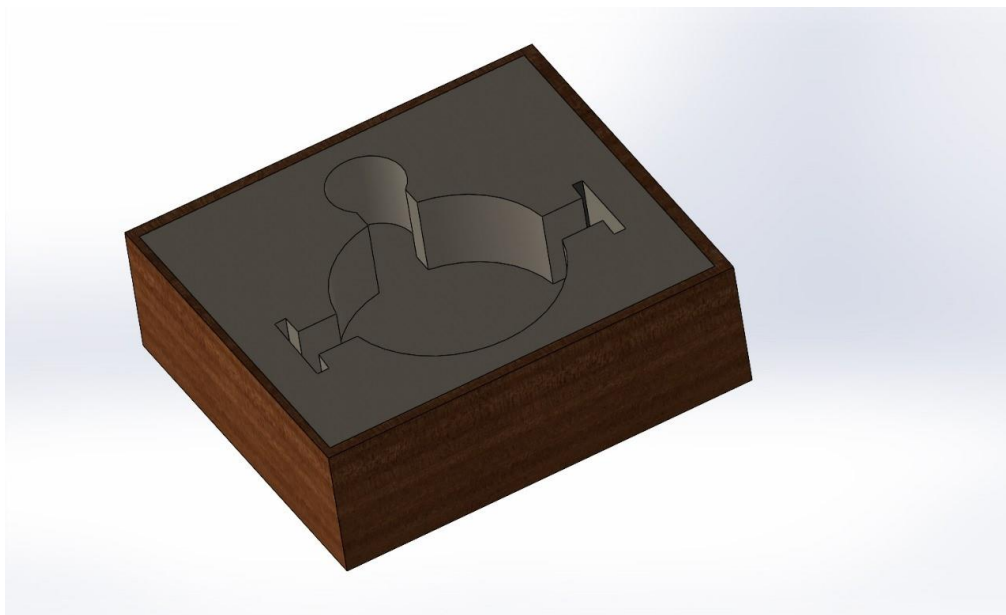


Figure 6: Drag for sand casting

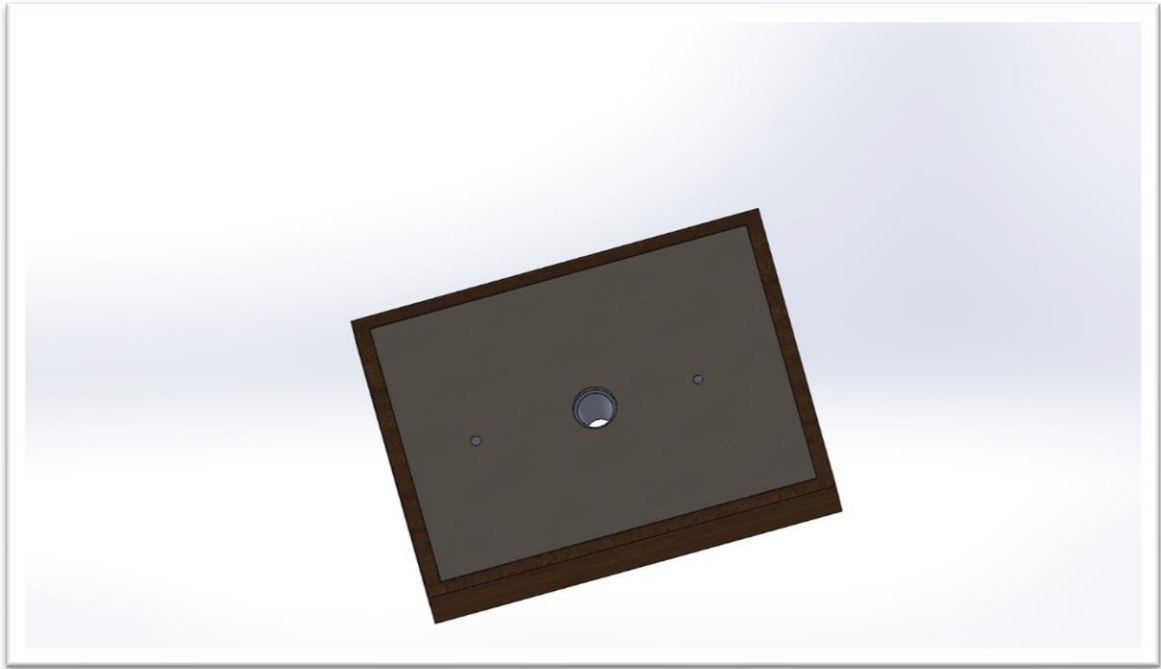


Figure 7: Cope for sand casting

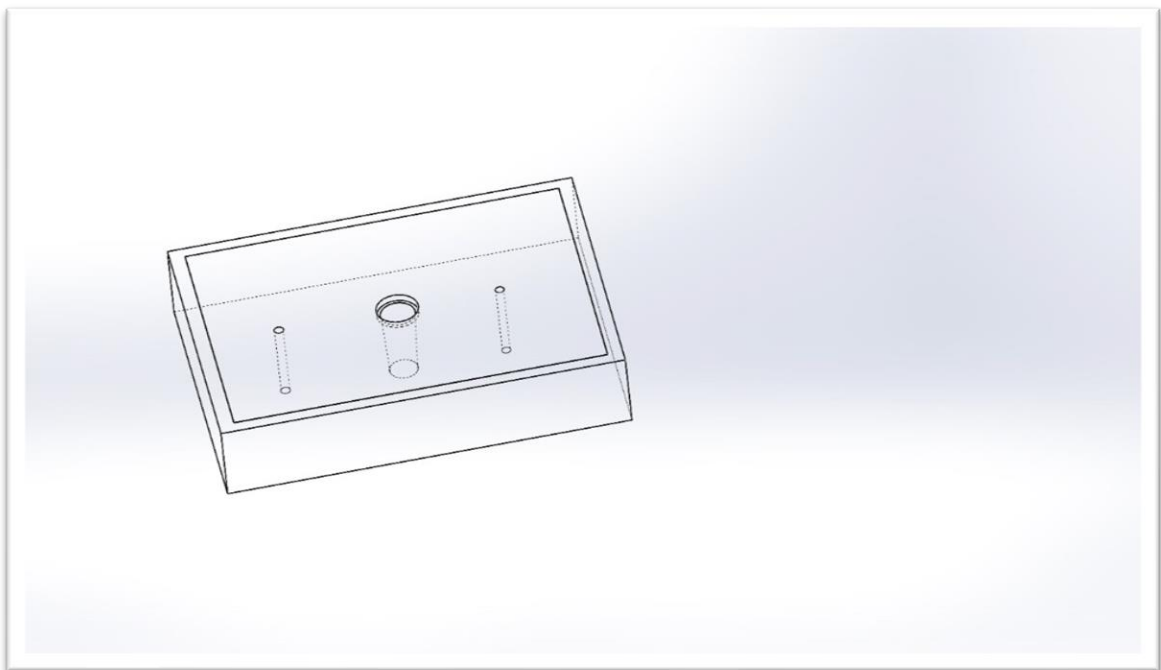


Figure 8: Wireframe view of cope

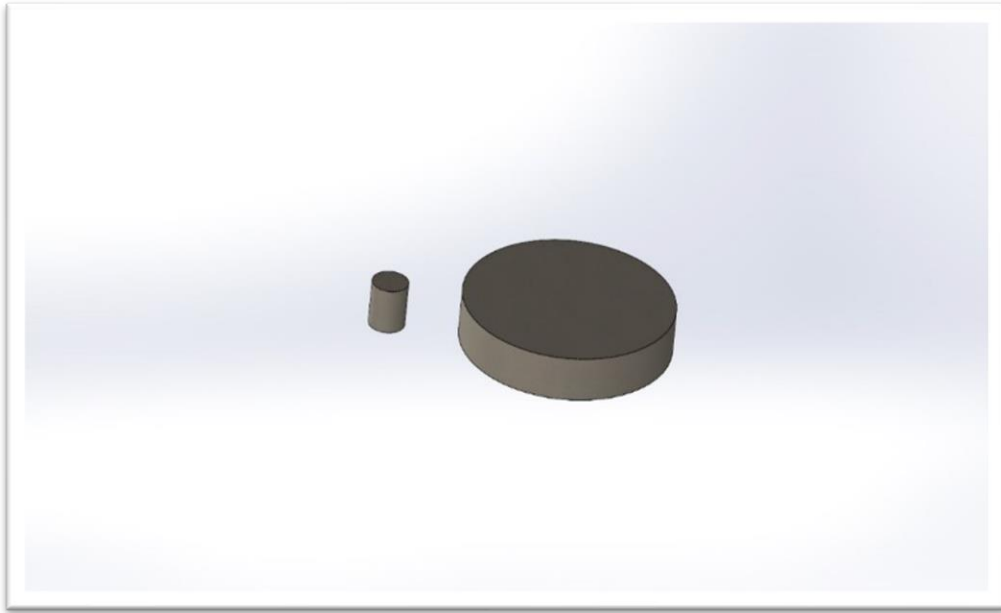


Figure 9: Mold cores



Figure 10: Mold pattern

*Each figure in this report was obtained from its respective Solid Work[®] 3D model. Access for those 3D models were provided in the “Appendices” part of this report.

Handle

Manufacturing the handle through casting is technically feasible, but it might not be the most optimal method as in pump barrel. Here are the reasons:

1. Material Properties: Medium carbon steel is typically more suited for forging rather than casting due to its composition and desired mechanical properties.
2. Complexity and Precision: because the thickness of the handle is low, it might be challenging to achieve such dimensions from casting process.
3. Cost and Efficiency: extra cost can be applied to finishing processes.

Possible changes to the material properties due to the moulding process

1. Casting typically yields a coarser grain structure compared to forging, impacting steel's mechanical properties like strength, toughness, and ductility.
2. Trapped gases during casting can create porosity within the material, acting as stress concentration points and diminishing the handle's overall strength and integrity.
3. Inclusions of oxides and impurities introduced during casting can weaken the material, negatively affecting its mechanical properties.
4. Cooling rates and solidification processes in casting can induce residual stresses, potentially causing warping or cracking and affecting dimensional stability and mechanical behavior.
5. The surface finish of a cast handle may be rougher and less uniform than other manufacturing methods, affecting aesthetics and potentially requiring additional finishing processes for desired smoothness.
6. Heat treatment processes like annealing or quenching may have limited effectiveness for cast components due to uneven cooling rates inherent in casting, impacting the material's microstructure and properties non-uniformly.

Possible casting processes and the reasons for the selection

1. Investment Casting
2. Sand Casting
3. Die Casting
4. Lost Foam Casting

The most suitable casting process for the handle is investment casting. Investment casting is chosen because it offers the ability to produce intricate and detailed components with high dimensional accuracy and excellent surface finish, which is essential for a handle that requires precise ergonomic features and comfortable grip for the user. Additionally, investment casting allows for flexibility in design variations and can accommodate the specific requirements of the handle design while providing good mechanical properties for medium carbon steel. As in the pump barrel, casting will not be the most suitable manufacturing method for produce this component.

4 Task C - Discussing the forming possibilities

Pump Barrel

Manufacturing a metallic pump barrel through metal forming is feasible. Metal forming processes like forging, extrusion, or drawing can be utilized for this purpose. Here are some reasons:

1. 1.Material Properties: Medium carbon steel possesses favorable properties for metal forming processes such as good formability, strength, and wear resistance, making it suitable for creating pump barrels by forming process.
2. 2.Dimensions and shape: Metal forming allows to create low thickness shapes.
3. 3.Strength and Durability: Metal forming processes can impart strength and durability to the pump barrel.
4. 4.Cost-effectiveness: Metal forming techniques can often be more cost-effective compared to other manufacturing methods, especially for producing large quantities of components like pump barrels.

Possible forming processes and the reasons for the selection

1. Forging
2. Extrusion
3. Rolling

The most suitable forming process for the pump barrel made of medium carbon steel for a handheld bicycle pump is extrusion. Extrusion is chosen because it allows for the creation of long, uniform shapes with high precision. In the case of a pump barrel, extrusion can produce the cylindrical shape with consistent wall thickness required for efficient operation. Additionally, extrusion provides excellent control over the material's grain structure and mechanical properties, resulting in a strong and durable pump barrel.

Base

Manufacturing a grey cast iron base for a handheld bicycle pump through metal forming is not feasible. Grey cast iron is typically manufactured using casting processes like sand casting or investment casting due to its brittleness and inability to deform plastically. Metal forming processes such as forging or stamping are more suitable for ductile materials like steel or aluminum, which can undergo plastic deformation without fracturing. Attempting metal forming with grey cast iron would likely result in cracking or other defects due to its lack of ductility.

Possible forming processes and the reasons for the selection

The most suitable forming process for the production of a metal base made of Gray cast iron for a handheld bicycle pump is Forging.

Forging involves shaping the metal by applying compressive forces using hammering, pressing, or rolling. It is highly versatile and suitable for creating solid components with various shapes and sizes. Forging offers excellent control over the grain structure of the material, resulting in improved mechanical properties such as strength, toughness, and fatigue resistance. Additionally, forging allows for the creation of complex shapes and features, making it an optimal choice for producing the metal base with intricate designs and precise dimensions.

Handle

Metallic handles for handheld bicycle pumps can indeed be manufactured through metal forming processes. Metal forming techniques such as forging and machining can be used to shape the handle from medium carbon steel. Here are a few reasons why metal forming is suitable for this application:

1. **Strength and Durability:** Medium carbon steel offers excellent strength and durability, and the metal forming processes can help maintain these properties by preserving the integrity of the steel's microstructure.
2. **Cost-effectiveness:** Metal forming processes can often be more cost-effective for manufacture components like handle when compared to the finishing processes included when casting.
3. **Material Utilization:** Metal forming processes typically generate less waste compared to subtractive manufacturing methods like machining.
4. **Surface Finish:** Metal forming processes gives a better surface finish than that in casting process.

Possible forming processes and the reasons for the selection

1. Forging
2. Extrusion
3. Rolling
4. Rotary swaging

The most suitable forming process for handle is rotary swaging. This process is chosen because it offers precise control over the shaping of the metal, allowing for the production of handles with required designs and precise dimensions. Rotary swaging also provides uniform deformation along the length of the component. Additionally, rotary swaging is well-suited for high-volume production, making it an efficient choice for manufacturing handles in large quantities.

Volume forming and sheet metal forming methods

Selected volume forming method – **Forging**

For the manufacturing of metal base of the handheld bicycle air pump, we choose volume forming process as a alternative method to manufacture even if the suitable process is casting. In volume forming process, there are many tools are used to form the metal block into the desired shape. Some of them are listed below.



Figure 11: Sledgehammer

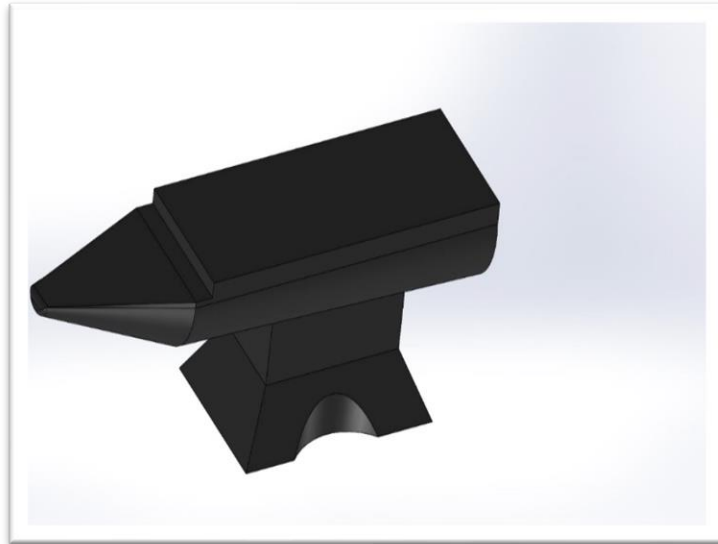


Figure 12: Anvil

Selected Sheet metal forming method – **Rolling**

Rolling is a metal forming technique where metal stock is passed through rotating rolls to reduce thickness or alter shape. It's widely used in various industries for mass-producing precise metal component as well as to produce metal blocks and metal sheets which are useful in various applications. In rolling, compressive forces exerted on the metal, leading to plastic deformation. Rolling produces sheets, plates, bars, and structural shapes efficiently and cost-effectively.

The main tool that is used in rolling is rolling mill which is similar to sheet metal forming machine.

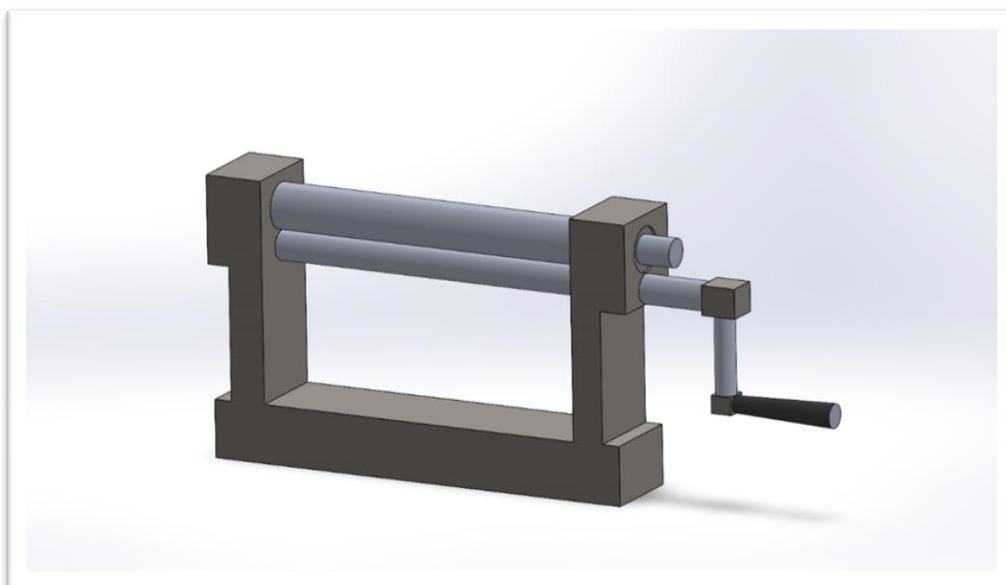


Figure 13: Sheet metal forming machine (rolling mill)

5 Task D – Non-metallic manufacturing processes

Nozzle

A polypropylene nozzle can be manufactured through plastic shaping or forming processes. Polypropylene, being a thermoplastic polymer, is conducive to various manufacturing techniques such as injection moulding, extrusion, or blow moulding. Here are the reasons:

1. **Injection Moulding:** This process involves injecting molten polypropylene into a mold cavity. It's highly efficient for mass production, offering precise control over nozzle dimensions and features.
2. **Extrusion:** Polypropylene can be extruded through a die to create continuous shapes with uniform cross-sections. While this method might be more suitable for simpler nozzle designs, it's cost-effective for large-scale production.
3. **Blow Moulding:** Suitable for hollow or tubular shapes, blow moulding uses compressed air to expand molten polypropylene against a mold surface. It's ideal for producing lightweight and durable nozzles.

Possible metal/plastic/ rubber shaping or forming processes and the reasons for the selection

1. Injection moulding
2. Blow moulding.
3. Thermoforming
4. Extrusion
5. Composite manufacturing

The most suitable process for manufacturing a polypropylene nozzle for a handheld bicycle air pump is injection moulding because of its high production rate, dimension accuracy and cost-effectiveness.

Exhaust Hose

We can manufacture a thermoplastic polyurethane hose pump through plastic shaping or forming methods. Thermoplastic polyurethane (TPU) is a versatile material that can be molded or extruded into various shapes, including hoses. Here are reasons why:

1. **Versatility:** TPU can be molded or extruded using various plastic shaping techniques, such as injection moulding, extrusion moulding, or blow moulding, to create complex shapes like hoses.
2. **Durability:** TPU offers excellent durability and flexibility, making it suitable for applications like hoses in bicycle pumps where resilience to bending and stretching is required.
3. **Chemical Resistance:** TPU is resistant to many chemicals, oils, and abrasions, making it suitable for outdoor and demanding applications like bicycle pump hoses.
4. **Temperature Resistance:** TPU can withstand a wide range of temperatures, ensuring that the hose remains functional even under varying weather conditions.

Possible metal/plastic/ rubber shaping or forming processes and the reasons for the selection

1. Extrusion
2. Injection moulding (for certain components of the hose)
3. Blow moulding (for certain components of the hose)
4. Compression moulding (for specific shapes or components)

The most suitable process for manufacturing a hose made of thermoplastic polyurethane is extrusion. This process allows for continuous production of hoses with consistent dimensions and properties, making it ideal for high-volume manufacturing. Extrusion can produce hoses of various lengths and diameters, and it offers flexibility in incorporating reinforcement layers or other features to enhance durability and performance. Additionally, extrusion is a cost-effective method for producing hoses with complex shapes and configurations required for air pump applications.



Figure 14: Extrusion machine

Image source: <https://th.bing.com/th/id/OIP.w9mimyOpMetm-gWtPsdjmQHaff?rs=1&pid=ImgDetMain>

6 Task E - Product assembly

Assembly processes which've been used in the order.

1. Attach the aluminium alloy pedals securely to the metal base using mechanical fastening techniques such as bolts or screws.
2. Secure the pump barrel firmly to the metal base using adhesive bonding to ensure a strong and durable connection.
3. Insert the exhaust tip and hose into designated slots on the metal base, utilizing snap-fit assembly for a quick and reliable attachment.
4. Combine the nozzle components and connect them to the hose, employing a press-fit assembly method for a snug and secure fit.
5. Assemble the piston and piston rod, then carefully insert them into the pump barrel, utilizing friction fit to ensure smooth operation.
6. Seal the pump barrel by attaching the lid and washer, employing a threaded connection for a tight and leak-proof seal.
7. Affix the handle to the piston rod using a combination of mechanical fastening and snap-fit assembly, ensuring a sturdy and ergonomic grip for the user.

Component	Assembly method	Reasons.
Pump barrel	Adhesive bonding Thread	to ensure a strong and durable connection.
Base	Mechanical fastening	Alignment and precision. Adaptability and cost-effective.
Nozzle	Press-fit assembly Rivets	Stability and strength.
Hose	Snap-fit assembly	Ease of assembly Flexibility.
Handle	Mechanical fastening Snap-fit assembly	Convenient and efficient in implementation.

Exploded view

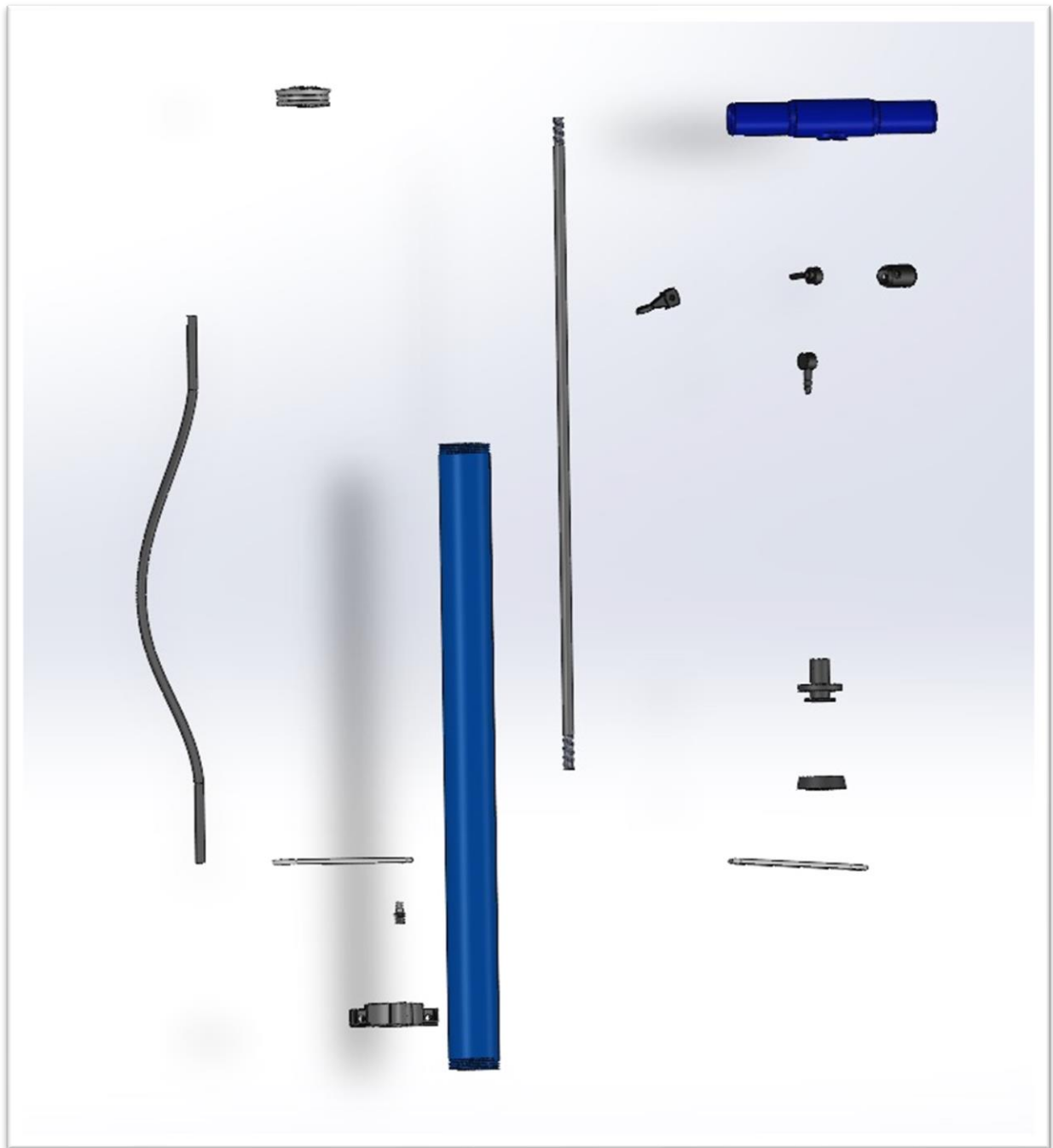


Figure 15: Exploded view of the air pump

*Link for the exploded view video clip: <https://drive.google.com/file/d/1f16ieg8hqxg5-qm3UwwEcAiJnyBKSNUw/view?usp=sharing>

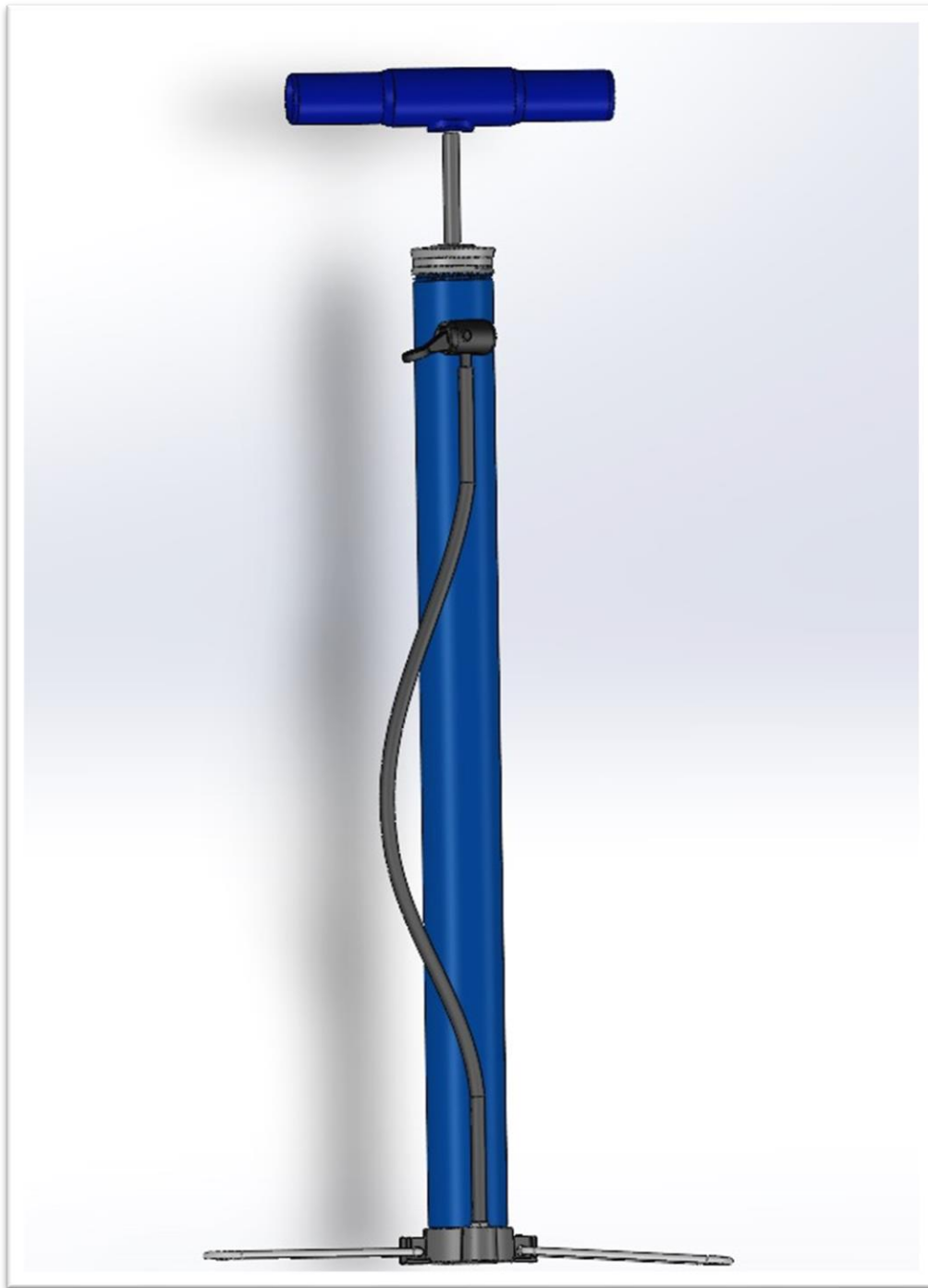


Figure 16: Final assembly

7 Conclusions

In conclusion, this reverse engineering project provided a comprehensive analysis of a handheld bicycle air pump. Through material identification, manufacturing process examination, and 3D modelling, we gained a deeper understanding of the pump's design and functionality. The project also identified areas for improvement, such as increased efficiency and multi-functionality. By considering casting and metal forming possibilities, we can explore potential future iterations that enhance the pump's performance and user experience. This project demonstrates the value of reverse engineering in not only understanding existing mechanisms but also sparking innovation for future development.

8 References

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Appendices

Solid Work files of the drawing in the report:

<https://drive.google.com/drive/folders/1dVgGWlZJoedc7FCKN54LGvYB7JzylQu2?usp=sharing>